

Buice 2D Diffuser

Introduction

This validation study examines the separated flow through a 2D asymmetric diffuser using the Menter SST, Chien k-e, and Spalart-Allmaras turbulence models. It was completed in order to investigate WIND's ability to predict the correct separation behaviors with smooth walls and adverse pressure gradients. The inlet flow is two-dimensional, incompressible, turbulent, and fully-developed channel flow with a Reynolds number of 20,000 based on the centerline velocity and the channel height. The prediction of the separation and reattachment points and the extent of the recirculation in these types of problems is particularly challenging for computational models. This flow has been experimentally studied at least by two different research groups: Obi *et al* at Keio University in Japan, and Buice & Eaton at Stanford University. In this study, numerical results are compared with experimental data in the form of axial velocity plots and skin friction data.

Please Note: This webpage has been optimized to be viewed with Microsoft Internet Explorer.

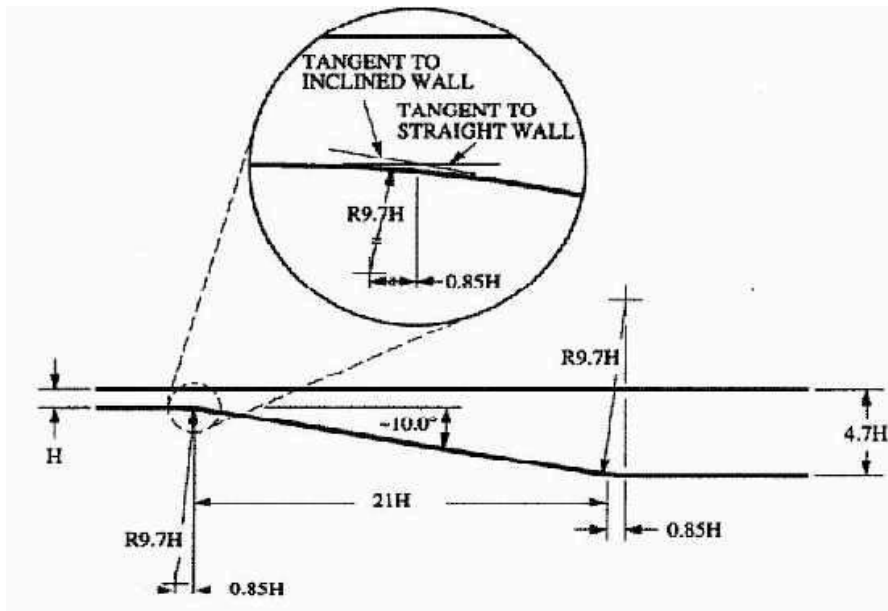


Figure 1. The geometrical description of the diffuser from Buice and Eaton.

Download tar File

The runtime files for the SST model case are contained in the Unix compressed tar file [buice.tar.gz](#). The files can then be extracted by the unix command

```
uncompress -c buice.tar.gz | tar xvof -
```

The files buice.x, buice.cgd, buice.cfl and buice.dat are the grid file in plot3d format, grid file in WIND common file format, WIND solution file, and WIND input file, respectively. The original experimental data points can be found [here](#).

Grid and Diffuser Geometry

The 2D grid is contained in a PLOT3D grid file (unformatted, single-zone, whole, three-dimensional) named buice.x. It consists a single zone with grid dimensions 473 x 141 (Figure 2 left). The x- and y-axes are taken in the streamwise and straight-wall normal directions, respectively. The origin of the x-axis is located at the intersection of the tangents to the straight and inclined wall, as shown in Figure 1. The y-axis originates from the bottom wall of the downstream channel. The I1 surface is the inflow. The IMAX surface is the outflow. The J1 surface is the diffuser. The JMAX surface is the upper viscous wall. The common grid file is buice.cgd. Walls were clustered such that average y^+ values were 1. The height H of the inflow domain is 0.591 inches. The grid was created with Pointwise, Inc's GRIDGEN software. The inflow region is 110H long. The lower wall of the diffuser section is at $y = 0$, and the section is 4.7H high and extends 56H downstream (of the ramp). Grid sequencing was used at 2 levels (118x35) and 1 level (236x70) to get an initial solution. Solutions obtained on a 1st level grid sequencing were very similar to solutions obtained on the full grid. All solutions presented here are with the full, unsequenced grid. A comparison of skin friction along the lower wall for the three grid levels using the SST model can be seen in Figure 2 (right). From this, it can be seen that the fine grid is sufficient to capture the flow effects especially in the diffuser region ($x/H > 0$).

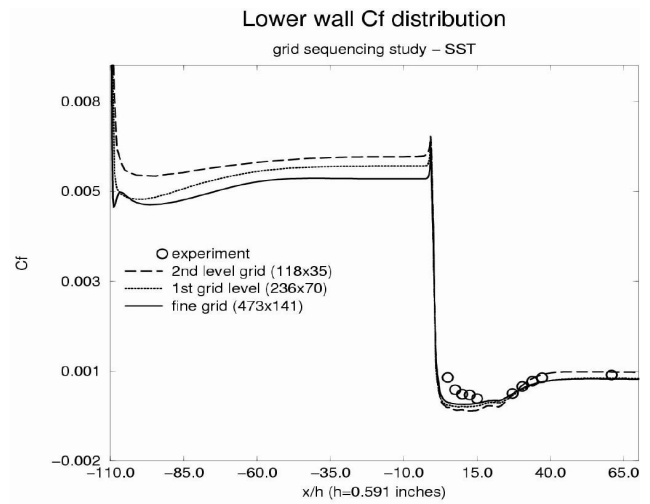
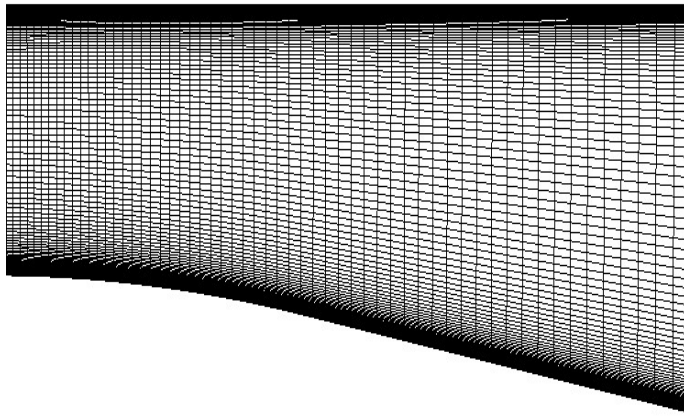


Figure 2. Zoomed view of computational grid (left) and grid sequencing study for three grid levels (right).

Initial Conditions

The initial flow conditions are simply the conditions of the **freestream** keyword as presented in Table 1.

Table 1. Static Freestream conditions.

Mach	Pressure (psia)	Temperature (R)	Angle-of-Attack (deg)	Angle-of-Sideslip (deg)
0.06	14.7	530.0	0.0	0.0

The bulk flow velocity U_b in the channel is 65 ft/sec, and the upstream channel centerline velocity is $1.14 U_b$ ($=74.1$ ft/sec). The static pressure was specified at the outflow BC (back pressure), and was adjusted slightly in order to obtain the correct U_b just upstream of the diffuser. The final pressure value for the SST model was 14.669 psi. This back pressure value was modified during the run cycle in order to obtain the same bulk flow velocity as the experiment just upstream of the diffuser. Another run was performed to investigate WIND's *massflow* specification option, and in order to conserve mass, WIND calculated and used a pressure of 14.6801 psi at the exit plane. About a 0.042 psi increase in back pressure was required for the SST model (compared with the S-A model). Assuming a static inflow temperature and pressure for normal wind tunnel conditions, and using the given Reynolds number of 20,000 and channel height H , the inlet Mach number (or flow velocity) of 0.06 was extrapolated and used to initialize the solution. From this, the bulk velocity *just upstream of the diffuser* was fine-tuned to match experimental conditions by slightly adjusting the back pressure (exit static pressure) while keeping the inflow static temperature and pressure specified in Table 1. The inflow conditions set with the *arbitrary inflow* keyword were the same as the reference conditions set with the *freestream* keyword.

Boundary Conditions

The I1 inflow boundary is specified with a **ARBITRARY INFLOW** boundary condition (BC). Static values with “hold totals” were specified uniformly across the inflow plane. The IMAX boundary is specified with an **OUTFLOW** boundary condition. The J1 and JMAX boundaries are specified with an **VISCOUS WALL** boundary condition. These are set in GRIDGEN. This problem was initially tried with a freestream BC specifying both total and static values (using hold totals) which presented unacceptable results and sluggish convergence. Profiles upstream of the diffuser were correct but axial velocity profiles in the diffuser itself were not matching experimental data (although, qualitatively, the solution looked correct). Using the *arbitrary inflow* BC corrected this problem. It is believed that the **FREESTREAM** BC only works for inflow boundaries where the flow conditions are uniform and do not change a lot from one point to another. For airfoil calculations, the freestream conditions are far enough away from the airfoil itself that the airfoil does not affect the freestream flow. At the inflow plane of a diffuser or nozzle, flow downstream in the nozzle will affect the freestream boundary (for subsonic flow), so this BC is not valid in this case. Using it will introduce errors downstream. The **ARBITRARY INFLOW** BC handles this situation properly. As a corollary exercise to understand this better, the Buice diffuser was run inviscid in the form of two sub-cases, one utilizing the *Freestream* BC and one using the *Arbitrary Inflow* BC. Both solutions gave very similar results. This confirms that having uniform conditions at the freestream should give similar results to using the *Arbitrary inflow* BC (for duct or channel flows).

Computational Strategy

The computation is performed using the time-marching capabilities of WIND to march to a steady-state (time asymptotic) solution. Local time stepping is used at each iteration. The time-marching is performed until convergence criteria is achieved. The solution was started on a sequenced grid with the Spalart-Allmaras turbulence model and switched to the Menter SST and Chien k- ϵ models after an eddy viscosity field was developed (after about 20,000 iterations). It was noticed in earlier runs that mostly laminar boundary layers resulted if the solution was started from scratch with the SST model (not enough eddy viscosity in the flowfield). Upon restart, the SST and k- ϵ models use the eddy viscosity values from the S-A model, resulting in a more realistic level of turbulence in the flowfield. Default turbulence model options were used with the SST and k- ϵ models in this study.

Input Parameters and Files

The WIND input data file for this case is buice.dat. The **arbitrary inflow** keyword indicates that the static flow conditions are specified as Mach number, pressure (psia), temperature (R), angle-of-attack (degrees), and angle-of-sideslip (degrees). Total pressure and total temperature are held constant at the inflow boundary. The **turbulence model** keyword indicates that the SST turbulence model is to be used. The **cycles** keyword indicates that a maximum of 20000 cycles will be run. The **iterations per cycle** keyword indicates that 5 iterations will be run per cycle. The **cfl#** keyword indicates that a CFL number of 2.0 is used. By default, WIND uses local maximum allowable time-step based on the specified CFL number. The **converge level** keyword indicates that the computation will stop if the L2 norm of the solution drops to 1.0E-09.

Post-Processing

The CFPOST utility was used to get [axial velocity profiles](#) across the diffuser. The bulk velocity is used to normalize certain quantities. It was obtained by integrating the axial velocities across all vertical points at the inflow plane and then dividing by the total distance. The skin friction values were obtained by finding the dynamic viscosity and slope of the axial velocity at the wall (du/dy), and then normalizing by the freestream density and bulk velocity.

Comparisons of the Results

Axial velocity contours (u-component of velocity) for the SST solution are shown in Figure 3. The S-A and k-e solutions gave similar qualitative results and are not shown here. Mean velocity profiles vs experimental data are shown in Figure 4 for different stations downstream (left to right) for the SST, S-A with curvature correction ON, S-A with curvature correction OFF, and k-e models. Both axis are normalized by the inflow height H (0.591 inches). Curve #1 (to the far left in each of the images) represents the fully developed channel flow. The downward slope of the diffuser ramp starts at x/H of 0. The separation region can be seen in Figure 5.

Figure 6 shows the lower and upper wall skin friction vs experimental data on the fine grid for all models. Analysis of the WIND's skin friction coefficient indicates that separation starts at an x/H of 1.9168, and the flow appears to reattach at an x/H of 30.67934 for the SST model. Separation starts at an x/H of 4.0377, and the flow appears to reattach at an x/H of 32.680 with the Spalart-Allmaras rotation correction OFF. Separation starts at an x/H of 3.6051, and the flow appears to reattach at an x/H of 33.044 with the Spalart-Allmaras rotation correction ON. The S-A solution with the curvature correction ON is not significantly different than with it OFF, other than the correction slightly improves the solution on the upper wall near an x/H of 0. For the Chien k-e model, separation starts at an x/H of 0.7583, and the flow appears to reattach at an x/H of 20.943. The S-A solutions produce a slightly higher C_f value (compared with the SST solution) along the upper wall of the diffuser. It underpredicts C_f at x/H of 0 compared with the SST model.

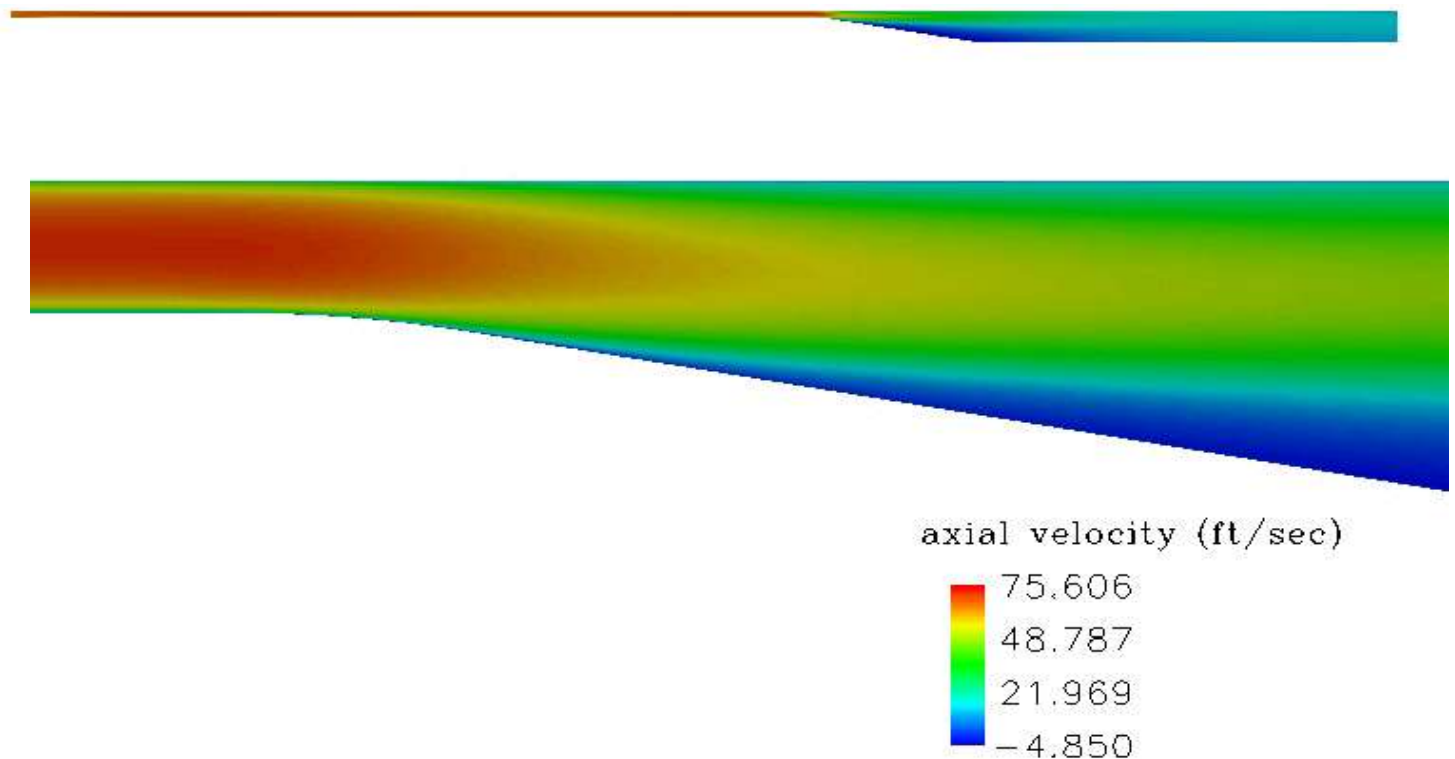
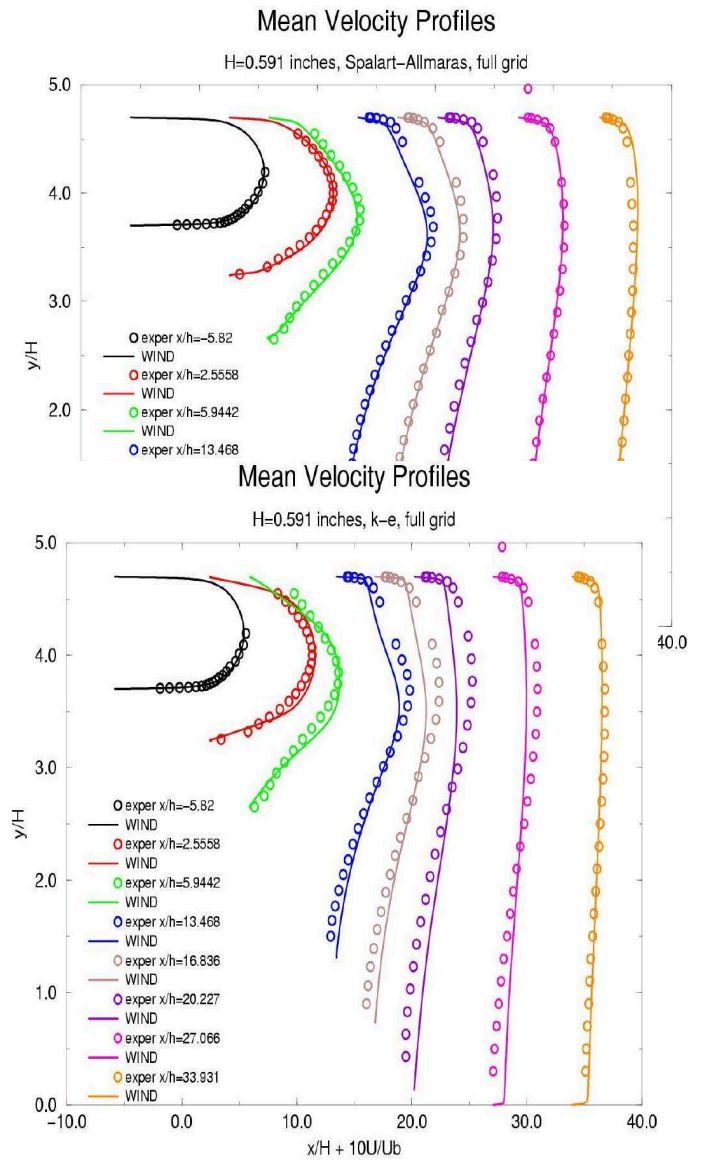
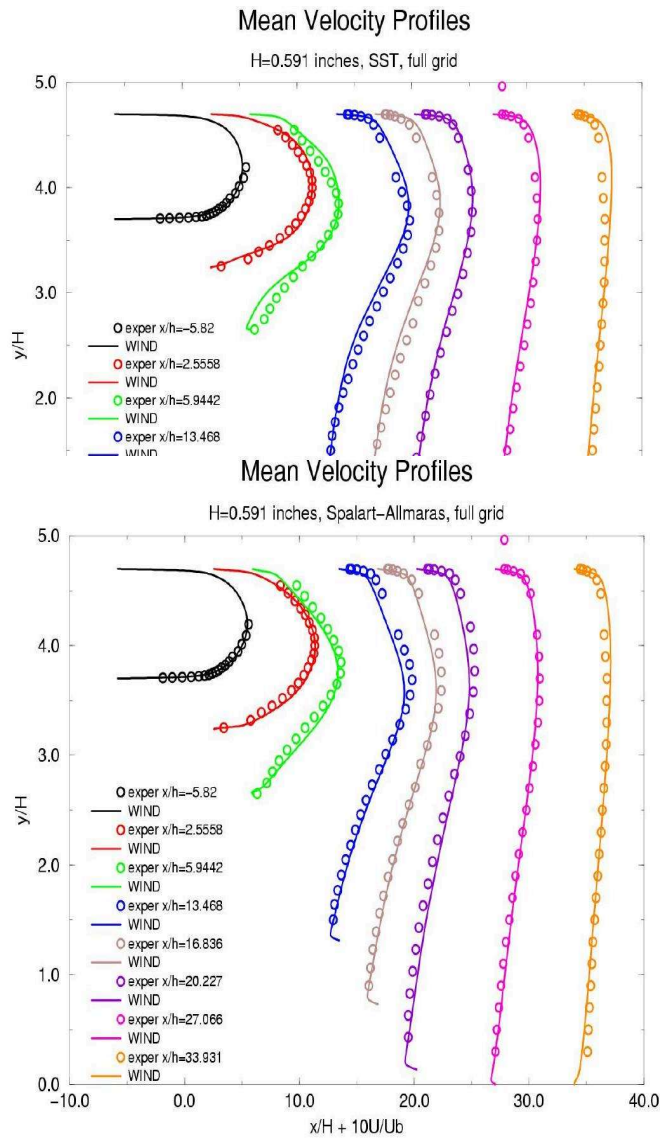
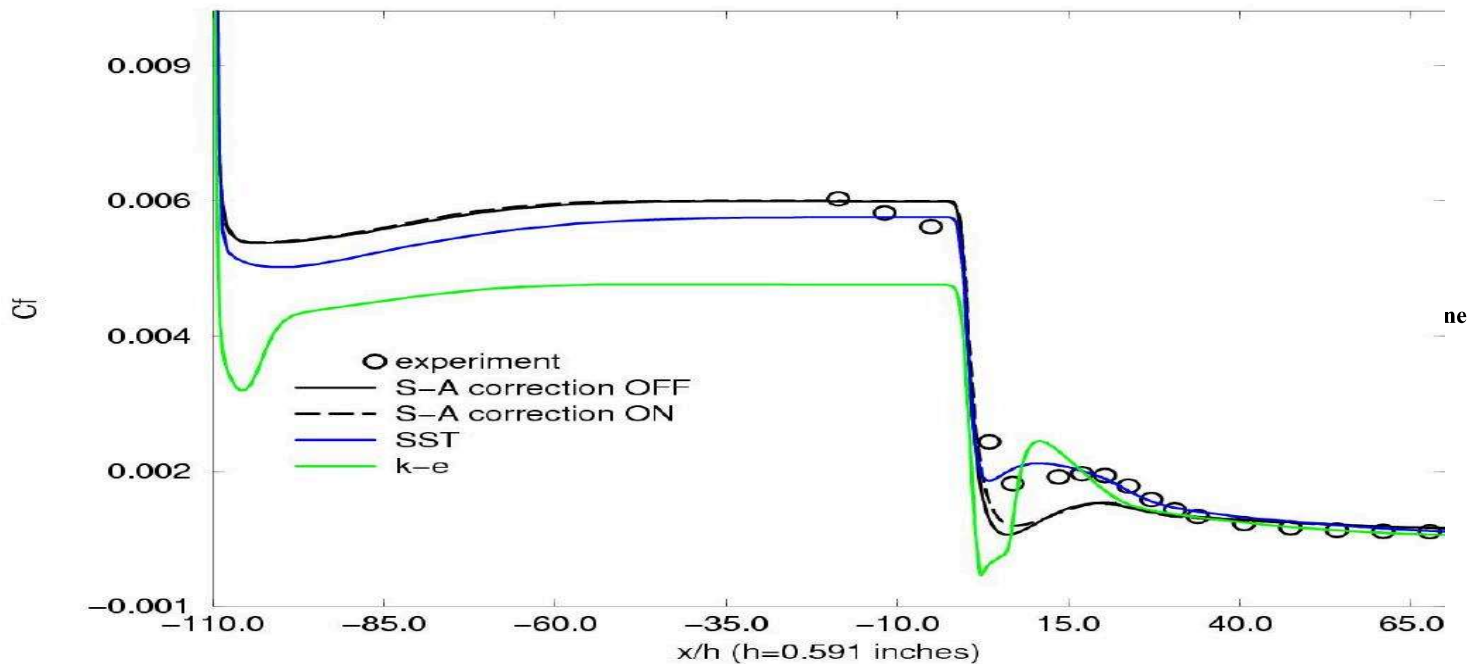
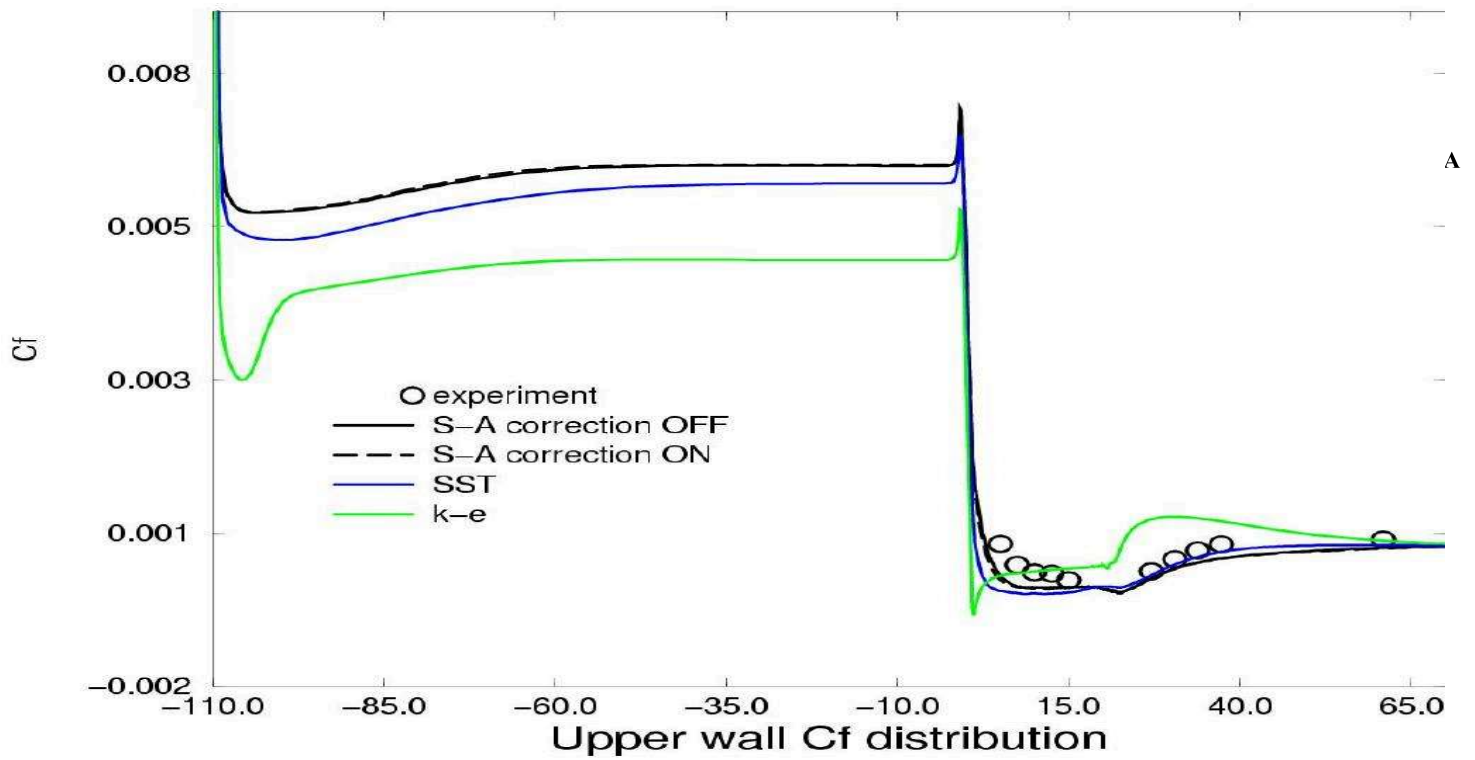


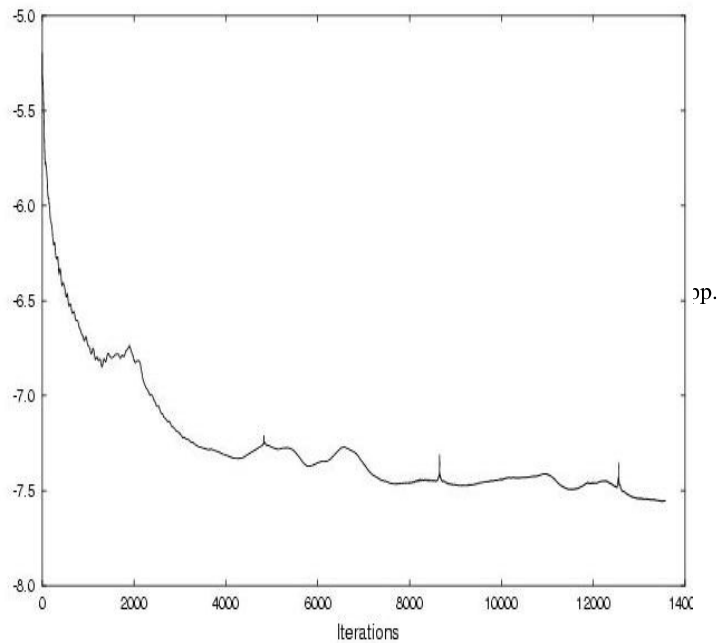
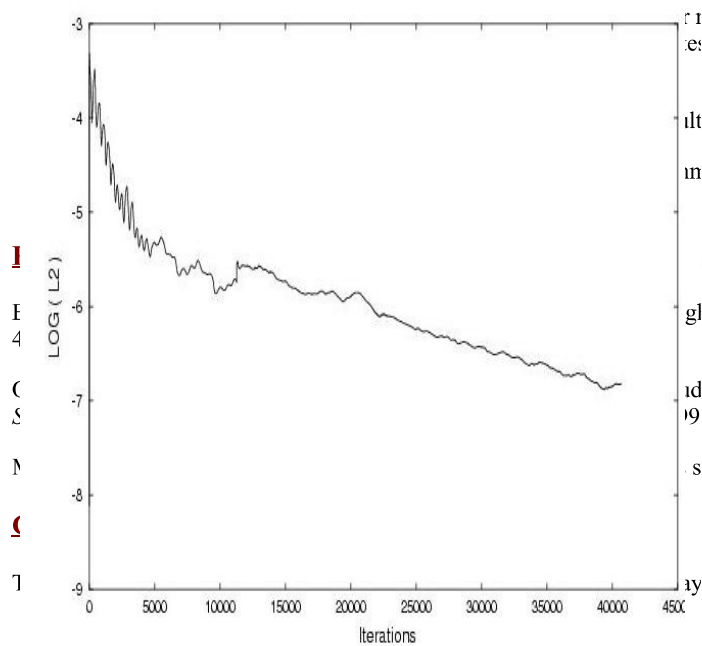
Figure 3. Axial velocity contours (zoomed and unzoomed) for the Buice subsonic 2D diffuser with the SST model. The first image is unzoomed showing the entire computational domain and the second is zoomed near the diffuser ramp area highlighting the separation region.



Lower wall Cf distribution



1. This solution would not develop properly if the inflow plane boundary condition was "freestream". Using instead the "arbitrary inflow" produced reasonable results. This was so even when specifying identical inflow conditions. Non-uniform flow properties across the inflow are believed to contaminate the solution when using the Freestream BC for duct flows.
2. Starting the solution from scratch using the SST model resulted in a very laminar flowfield resulting in too little separation inside the diffuser. This has to do with how SST initializes the flowfield when it starts from scratch. Starting the solution with the S-A model and switching to the SST model solved this problem.
3. The backpressure (outflow pressure) required at the exit (in order to get nearly the same flow speed just upstream of the diffuser) is dependent on the turbulence model used. The SST model required a higher backpressure than the S-A model. This means that for the same backpressure, the S-A model produced higher flowspeeds through the channel (prior to entering the diffuser).
4. Qualitative and quantitative results for the S-A and SST models were very similar overall. However, the SST model better predicted the skin friction distribution on the upper wall. The k-e model showed the largest deviations from experiment.



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